Title Title Title

Extended Essay

EC423 — Labour Economics

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Abstract

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Todo:（按先后优先级）

- General DiD in Empirical Analysis
- Literature Review
- Theoretical Model
- Conclusion
- Discussion

1 Introduction

Since the launch of ChatGPT in autumn 2022, generative AI has undergone rapid development over the past three years. Beyond text generation, AI has achieved notable breakthroughs across a range of creative domains — including code generation, image synthesis, music composition, and video production. The pace of these advances has far exceeded the expectations of most observers, raising pressing questions about the implications for workers in occupations traditionally considered resistant to automation.

Early empirical evidence suggests that these advances are already reshaping labour markets for creative workers. Freelancers on Upwork in AI-exposed occupations experienced a 2 per cent decline in the number of contracts and a 5.2 per cent fall in earnings following the release of major generative AI tools, with image-related workers suffering considerably steeper losses (Hui et al., 2024).

A recent paper by Pabba (2026) argues that AI tools should not be treated merely as software, but rather as a form of capital.

Measuring the impact of AI is hard. Unlike discrete, large-scale shocks such as the COVID-19 pandemic, constructing a credible counterfactual for the impact of AI on the labour market is particularly challenging, precisely because AI adoption is gradual and diffuse rather than sudden (Massenkoff & McCrory, 2026).

Another focus about this essay is about unemployment. I want to use the QLFS data in UK to investigate the impact of AI on unemployment, and whether the impact is different for junior and senior workers within occupations.

2 Literature Review

3 Theoretical Framework

Consider a single occupation producing output Y, by two types of tasks:

- Task L: Low expertise, including routine jobs, easy be replaced by AI automation.
- Task H: High expertise, including context-dependent tasks, management tasks, and other tasks requiring accumulated experience.

The output Y follows a CES aggregation:

$$Y = \left[\beta y_L^{\frac{\sigma-1}{\sigma}} + (1-\beta) y_H^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}} \tag{1}$$

where σ is the within-occupation elasticity of substitution between the two types of tasks, which is distinct from the between-occupation elasticity in Autor & Thompson (2025), and β is the share of low-expertise tasks in the occupation.

4 Data and Identification Strategy

The data I used here is the positions dataset provided by Revelio Labs, which contains detailed information on job postings in the UK. For analysis, I used data from 2019 to 2025, which allows me to capture the period before and after the release of ChatGPT tool in late 2022.

For each position, I observe the predicted salary, the O*NET occupation code, and other basic characteristics. Two additional variables are particularly relevant to this study. The first is an ordinal seniority variable, ranging from 1 to 7, that captures

the hierarchical level of each position within an organisational structure. A detailed mapping is provided in Table 1.[1] I take this variable at face value throughout the analysis. Lichtinger & Hosseini Maasoum (2025) validate the measure by showing that the distribution of job-title keywords aligns closely with expected seniority distinctions and remains stable both before and after the diffusion of generative AI, confirming that it reliably captures positional hierarchy over time.

Seniority	Level	Examples
1	Entry	Accounting Intern, Software Engineer Trainee, Paralegal
2	Junior	Account Receivable Bookkeeper, Junior Software QA Engineer, Legal Adviser
3	Associate	Senior Tax Accountant, Lead Electrical Engineer, Attorney
4	Manager	Account Manager, Superintendent Engineer, Lead Lawyer
5	Director	Chief of Accountants, VP Network Engineering, Head of Legal
6	Executive	Managing Director (Treasury), Director of Engineering (Backend Systems), Attorney (Partner)
7	Senior Executive	CFO, COO, CEO

Table 1: Revelio Labs seniority classification

The second is the position number, which records the chronological order of each position within a given individual's profile, with lower values corresponding to earlier roles. This variable enables the analysis of within-individual career mobility, which I exploit in a later section.

I then merge the AI exposure scores constructed by Eloundou et al. (2024) to the positions dataset via the O*NET occupation code. Based on the merged scores, I classify occupations into three categories: low exposure (below the 25th percentile), medium exposure (25th to 75th percentile), and high exposure (above the 75th percentile).[2]

4.1 Workers' Mobility

From the

[1]A detailed description of the seniority classification methodology is available at: https://www.data-dictionary.reveliolabs.com/methodology.html#seniority.

[2]The exposure scores are publicly available at: https://github.com/openai/GPTs-are-GPTs.

这个部分讨论一下 Identification Strategy 为草稿

Like the key research question from Heckman (1979), for those people who do not work, we couldn't observe wage. It is therefore necessary to address sample selection bias. In my essay, the intensive margin regression only look at those workers' outcome who successfully transit, but people who really transit isn't necessarily random. Imagine of two junior workers at high AI exposed jobs, one is with high ability one is not. During the post-ChatGPT period, those with better ability would transit to higher seniority jobs, and those weaker would stay or even drop out of the labour market. This would lead to higher Δ Seniority, it is not because AI helps junior, it is because only "winners" are observed.[3]

5 Empirical Analysis

5.1 Career Mobility

In this section, I turn to the analysis of career mobility, with particular emphasis on the post-ChatGPT period. The individual-level panel structure of the Revelio Labs data allows me to track career transitions directly. This addresses a limitation explicitly acknowledged by both Lichtinger & Hosseini Maasoum (2025) and Klein Teeselink (2025), who identify the analysis of individual career trajectories following AI exposure as a direction for future research.

For each transition, I construct outcome variables capturing the change in seniority level, the log wage change ($\Delta \ln w$), and an indicator for whether the worker switched O*NET occupation code. The core specification throughout is a triple-difference model:

[3]The Heckman Model and the role of the inverse Mills ratio (Hazard ratio) in correcting for sample selection bias were covered in the EC423 Labour Economics lectures (Autumn Topic 1 and Autumn Topic 3), though the lecture didn't take a lot of part in Heckman two-step procedure. I apply the same logic here: the selection equation models entry into the transition sample, and the inverse Mills ratio corrects for non-random selection in the outcome equation. See Heckman (1979) for the original derivation.

$$Y_i = \beta_1 \text{Junior}_i + \beta_2 \text{AIExp}_i + \beta_3 \text{Post}_t + \beta_4 \text{Junior}_i \times \text{AIExp}_i + \beta_5 \text{Junior}_i \times \text{Post}_t + \beta_6 \text{AIExp}_i \times \text{Post}_t + \beta_7 \text{Junior}_i \times \text{AIExp}_i \times \text{Post}_t + \varepsilon_i \tag{2}$$

5.1.1 Extensive Margin: Who Moves?

Table 2 reports results from the extensive-margin analysis. Using the full sample of 27.5 million positions (including those without a subsequent transition), I estimate the triple-difference specification with an indicator for whether a transition is observed as the dependent variable. The estimate of β_6 is -0.373, indicating that senior workers in high-exposure occupations became substantially less mobile after ChatGPT, consistent with AI augmenting role-specific expertise and strengthening attachment to current positions. The triple-interaction term $\beta_7 = 0.228$ is positive and highly significant: relative to senior workers, junior workers in the same high-exposure occupations experienced a considerably smaller decline in transition probability. That is, while all workers in high-AI-exposure occupations became less mobile, this stabilisation was concentrated among senior workers; junior workers remained comparatively more exposed to turnover. The result is robust to excluding positions that began after July 2024 ($\beta_7 = 0.229$), ruling out mechanical right-censoring as a driver.

	Full sample	*Censoring-robust*
	(1)	(2)
Junior	0.194***	0.194***
	(0.038)	(0.038)
AI Exposure	0.365***	0.365***
	(0.053)	(0.053)
Post	0.222***	0.333***
	(0.031)	(0.029)
Junior × AI Exp.	–0.216***	–0.216***
	(0.053)	(0.053)
Junior × Post	–0.145***	–0.168***
	(0.030)	(0.028)
AI Exp. × Post	–0.373***	–0.352***
	(0.044)	(0.043)
Junior × AI Exp. × Post	**0.228*** **	**0.229*** **
	(0.042)	**(0.041)**
Constant	0.221***	0.221***
	(0.037)	(0.038)
N	27,472,206	24,531,424
R^2	0.009	0.022

Note: The dependent variable is an indicator for whether a position is followed by a transition to a different employer or occupation. Column (2) excludes positions with start dates after July 2024 to address potential right-censoring of ongoing positions. Cluster-robust standard errors in parentheses.

Table 2: Transition probability: triple-difference estimates

5.1.2 Intensive Margin: Transition Outcomes

Conditional on transitioning, Column (1) of Table 3 shows that junior workers leaving high-exposure occupations after ChatGPT gain significantly more seniority in their next position ($\beta_7 = 0.124$). However, this estimate is susceptible to positive selection: if AI disproportionately displaces weaker junior workers, those who successfully transition constitute a positively selected subsample, and the observed seniority gain overstates any genuine upskilling effect.

To address this concern, I implement a Heckman (1979) two-step correction. The selection equation models the probability of transitioning as a function of the main regressors and a firm-level exclusion restriction: the pre-period separation rate at the individual's company, which plausibly affects whether a worker transitions but not the characteristics of the destination position. The probit is estimated on a 10 per cent

random subsample for computational tractability, and the resulting inverse Mills ratio is included in the outcome equation estimated on the full sample.[4]

Column (2) reports the corrected estimates. The inverse Mills ratio is significant ($\hat{\lambda} = -0.174$), confirming that workers with lower predicted transition probability tend to experience smaller seniority gains conditional on transitioning — that is, positive selection into the transition sample is present. After correction, β_7 declines from 0.124 to 0.061, significant only at the 10 per cent level. This approximately 50 per cent reduction indicates that a substantial share of the observed seniority gain reflects compositional upgrading of the transition pool rather than a genuine treatment effect. Nevertheless, both interpretations reinforce the central thesis. If the effect is genuine, AI facilitates upward mobility for capable junior workers. If it reflects selection, AI filters out weaker junior workers from high-exposure occupations. Either channel implies that within-occupation composition shifts in the direction predicted by the model.

Columns (3) and (4) examine log wage changes upon transition. The triple-interaction term β_7 is essentially zero and statistically insignificant, indicating that junior workers are not differentially penalised in wage terms relative to senior workers. The adjustment mechanism thus operates primarily through the quantity margin — who remains in and who exits the occupation — rather than through differential wage compression.

Columns (5) and (6) examine occupational switching. The triple-interaction term is small and insignificant ($\beta_7 = -0.015$, $t = -1.00$ in the baseline). Junior workers do not systematically reallocate toward lower-AI-exposure occupations after ChatGPT. This null result is informative: the post-ChatGPT adjustment in career mobility operates along the seniority dimension rather than through occupational flight. Workers who transition move to positions of comparable AI exposure but at higher seniority — a

[4]The probit is estimated on observations where $\text{mod}(\text{user_id}, 10) = 0$. Coefficients are used to compute the linear index and the Mills ratio for all observations. Results are virtually identical when estimated on alternative subsamples ($\text{mod} = 1, 2, ...$).

pattern more consistent with upward filtering within related occupations than with wholesale retreat from AI-exposed work.

	Δ Seniority		*Δ ln Salary*		*Occ. Switch*	
	(1)	(2)	(3)	(4)	(5)	(6)
Junior	1.241***	1.197***	–0.042**	–0.042**	0.161***	0.180***
	(0.069)	(0.067)	(0.019)	(0.018)	(0.047)	(0.048)
AI Exposure	0.683***	0.591***	0.088***	0.087***	–0.143	–0.103
	(0.127)	(0.123)	(0.030)	(0.029)	(0.093)	(0.094)
Post	0.104***	0.063***	–0.032***	–0.033***	–0.004	0.013
	(0.017)	(0.018)	(0.009)	(0.008)	(0.009)	(0.009)
Junior × AI Exp.	–0.329***	–0.274**	–0.005	–0.004	–0.120*	–0.143**
	(0.109)	(0.107)	(0.030)	(0.029)	(0.072)	(0.073)
Junior × Post	–0.118***	–0.084***	0.039***	0.040***	–0.004	–0.019**
	(0.018)	(0.018)	(0.007)	(0.007)	(0.009)	(0.008)
AI Exp. × Post	0.052*	0.144***	–0.110***	–0.108***	0.019	–0.021
	(0.029)	(0.031)	(0.015)	(0.015)	(0.016)	(0.016)
Junior × AI Exp. × Post	**0.124*****	**0.061***	**–0.003**	**–0.004**	**–0.015**	**0.012**
	(0.034)	**(0.035)**	**(0.013)**	**(0.013)**	**(0.015)**	**(0.015)**
Δ Seniority			0.202***	0.202***		
			(0.003)	(0.003)		
IMR (λ)		–0.174***		–0.003		0.075***
		(0.019)		(0.005)		(0.011)
Constant	–0.801***	–0.612***	0.068***	0.071***	0.688***	0.606***
	(0.076)	(0.074)	(0.019)	(0.017)	(0.059)	(0.063)
N	12,487,869	12,487,869	10,676,912	10,676,912	12,487,869	12,487,869
R^2	0.138	0.139	0.341	0.341	0.020	0.022
Heckman correction	No	Yes	No	Yes	No	Yes

Table 3: Career mobility regressions: triple-difference estimates

6 Discussion

4.7: Discussion 部分仍然是想到哪写到哪，后期需要系统地进行整理和补充

Notice that research on the impact of AI on labour markets is still in its infancy, and there are many open questions about the mechanisms through which AI affects workers. One important argument is that we still couldn't find a consistent and convincing exposure measure for AI, which is crucial for empirical analysis. The most recent commonly used measurements can't broadly agree with each other on highly exposed

occupations, making it hard to draw robust conclusions about the impact of AI on labour markets (Gimbel et al., 2026).

6.1 Why Wage Prediction "Fails"?

7 Conclusion

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Appendix

A Validation from QLFS Data

I also use data set from Quarterly Labour Force Survey by UK government. And I am also aware that the dataset itself exists.